

hardware (logic gates, flip-flops, counters, etc), the functionality and circuit block interconnects. Various CAD tools are available to synthesise a circuit based on the HDL. The most widely used synthesis tools come from two CAD companies: Synposys and Cadence.

Without going into details, you may consider VHDL as 'C' of the VLSI industry. VHDL stands for 'VHSIC hardware definition language,' where VHSIC stands for 'very high-speed integrated circuit.' This language is used to design the circuits at a high level, in two ways. It can either be a behavioural description, which describes what the circuit is supposed to do, or a struc-

as operational amplifiers, filters or power management chips. For more complex analogue chips such as data converters, the design is done at the transistor level, building up to the cell level, then the block level and finally integrated at the chip level. Not many CAD tools are available for analogue design even today and thus analogue design remains a difficult art. SPICE remains the most useful simulation tool for analogue as well as digital design.

Where to apply skills?

Now, it's time to address the real-world need. Know where to apply your skillsets. For example, analogue

There are a number of directions you can take while choosing a career in VLSI, and they are closely related to each other. So be a go-getter, dig all the possibilities to gain a practical exposure.

tural description, which describes what the circuit is made of. There are other languages for describing circuits, such as Verilog, which work in a similar fashion. Both forms of description are then used to generate a very low-level description that actually spells out how all this is to be fabricated on the silicon chips. This results in the manufacture of the intended IC.

For an analogue design, the flow may vary to some extent, like specifications→architecture→circuit design→SPICE simulation→layout→parametric extraction/back annotation→final design→tape out to foundry.

While digital design is highly automated now, very small portion of analogue design can be automated. There is a hardware description language called 'AHDL' for analogue design but it is not widely used as it does not accurately give the behavioural model of the circuit because of the complexity of the analogue behaviour of the circuit. Many analogue chips are termed as 'flat' or non-hierarchical designs. This is true for small-transistor-count chips such

as designs are mostly used for small-transistor-count precision circuits such as amplifiers, data converters, filters, phase-locked loops and sensors.

In digital design, the progress in the fabrication of ICs has enabled designers to create fast and powerful circuits in smaller and smaller devices. This also means that one can pack a lot more of functionality into the same area. The biggest application of this ability is found in the design of application-specific integrated circuits (ASICs). These ICs are created for specific purposes—each device is created to do a particular job. The most common application area for an ASIC is DSP—signal filters, image compression, etc. To go to extremes, consider the fact that the digital wristwatch normally consists of a single IC doing all the time-keeping as well as extra features like games and calendar. Systems-on-a-chip (SoCs), on the other hand, are highly complex mixed-signal circuits (digital and analogue on the same chip). Network processor chips and wireless radio chips are examples of SoCs.

What's next?

It is possible that your awareness about most of the aforementioned terms is from a notional perspective only. But the industry needs something extra—actual hands-on experience is indispensable. Remember, there are a number of directions you can take while choosing a career in VLSI, and they are closely related to each other. So be a go-getter, dig all the possibilities to gain a practical exposure.

At the same time, be up-to-date about the latest technological trends to get an edge over your competitors. For example, reconfigurable computing is a very interesting development in microelectronics. It involves fabricating circuits that can be reprogrammed on the fly. Try to know how reconfigurable computing involves specially fabricated devices called FPGAs, that when programmed act just like normal electronic circuits without using a microcontroller running with EEPROM inside. This fantastic ability to create modifiable circuits again opens up new possibilities in microelectronics.

You may even explore how ASIC designing creates miniature devices that can do a lot of diverse functions. With an impending boom in this kind of technology, what the industry needs is a large number of people who can design these ICs. This is where one may cross the threshold between a chip designer and a systems designer at a higher level.

You will require extensive training in this field. If you have passion for chip designing and want to learn it, you need not disappoint. All IITs and other prestigious engineering institutes have included VLSI as an important part of their course curriculum.

So don't panic over the current job scenario, the time's perfect to resurrect your thought process. There's plenty of time at hand. Devote your time to become an expert in VLSI. It will definitely bring 'very large' opportunities in the future. ●

The author is a research analyst cum journalist at EBY